

An Ultra-Wideband Power Amplifier Based on GaN HEMT

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Abstract: The design, fabrication and measurements of an ultra-broadband power amplifier based on GaN HEMT technology which operates in the frequency ranging from 3 GHz to 8 GHz, is presented in this paper. A TGF2023-2-02 GaN HEMT chip from TriQuint is adopted. The amplifier is designed by using a frequency compensation and multi impedance broadband matching approach for both input and output networks. And a sector micro strip line is implemented in the input matching network to achieve the wideband higher gain features. The measured results show that the amplifier module provided more than 11 dB power gain over 3-8 GHz. The saturated output power is 37.3 dBm under DC bias of $V_{ds}=28$ V, $V_{gs}=-2.53$ V at the frequency of 5 GHz.

Keywords: GaN HEMT; ultra broadband; power amplifier; output power

1 Introduction

The power amplifier, as the most critical module in the transmitter, directly determines the overall performance of the system. Modern electronic warfare systems, personal and wireless communication, remote sensing, radar and aero spatial applications require electronic front-end systems with very high performance in terms of bandwidth, output power, and noise [1-2]. And the GaN HEMT shows clear advantages over other materials such as GaAs and Si, considering its characteristics such as wide band gap, and its potential in high output power density, especially the very high drain-to-source breakdown voltage, resulting in higher device's optimum impedance, simplifying the output transformation ratio towards 50-ohm and allowing broadband design [3]. Based on GaN devices, numerous amplifiers has been presented over a period of time by many researchers. However, the bandwidth of an amplifier whose output power is larger than 6 W is less than 3-8 GHz. Moreover, more distributed and balanced structures are adopted in broadband amplifiers, which both have a strong demand on the consistency of the power devices and simultaneously increase the size and cost of the circuits [4-8].

This paper proposes a wideband amplifier based on GaN HEMT consisting of a simple structure. The chip used in the circuit is unpackaged. Firstly, the bonding wire is modeled and simulated by HFSS (high frequency structure simulator). Secondly, a frequency

compensation and multi impedance broadband matching approach are implemented to achieve the wideband features including higher gain and output power, especially a sector micro strip line with broadband characteristics is used in the input matching network. The actual measurements demonstrate the availability and accuracy of the design approach. Good stability, high gain and better input & output return losses have been attained in the desired band. The proposed design offers more than 11 dB power gain over 3-8 GHz and that the output power is 37.3 dBm.

2 Simulation of bonding wire

The bonding wire is modeled and simulated by HFSS so as to accurately estimate the effect of the bonding wire for the circuit. Figure 1 is the model of the bonding wire in HFSS. No.1 and No.3 respectively represent ports of connecting bonding wires and the micro strips. No.2 and No.4 respectively represent ports of connecting bonding wires and the chip pads. The diameter of the bonding wire of gold is 25 μ m, the span is 400 μ m, the height is 100 μ m and the distance between is 260 μ m. Figure 2 shows the comparison of reflection coefficient $S(1, 1)$ of port1 in HFSS and ADS. It shows the larger range of the inductance in HFSS in the frequency from 3-8 GHz considering the actual pad geometry and material properties of transistor and PCB substrates. So the simulation results of HFSS are used for broadband amplifier design.

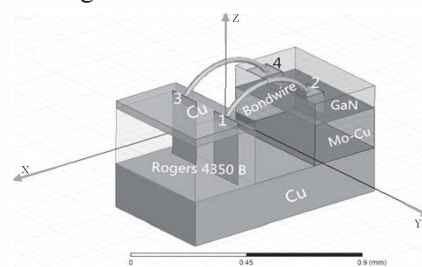


Figure 1 Model of bonding wire

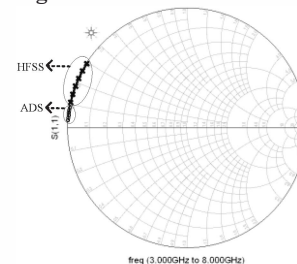


Figure 2 Simulation of bonding wire (from 3-8 GHz)

3 Broadband power amplifier design

Figure 3 shows the schematic diagram of the proposed wideband amplifier. This circuit consists of a matching circuit, DC-biased circuit and power manager. The DC bias and matching networks are all on the printed circuit board (PCB), which is a Rogers 4350 B with a higher dielectric constant of 3.66 which benefits for the small size of the matching components and a substrate thickness of 0.254 mm. Moreover, it is necessary to leave an appropriate hole for the chip to ensure the chip surface and micro strip line in the same plane. Both the input and output ports are connected to the standard 50 Ω load.

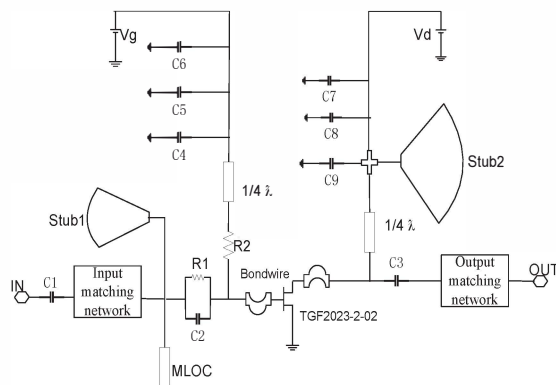


Figure 3 Schematic diagram of the proposed power amplifier

The output matching network is designed according to optimal loads found by the load-pull based on the model of the RF power transistor built before to ensure the optimal performance under large signal operations [9]. Based on these values, the optimum matching at frequency of 8 GHz for frequency compensation and multi impedance matching approach is applied for the broadband matching. Transmission line elements and capacitor in this matching network are needed to be tuned to reach optimized load matching in the whole frequency. In addition, the open radial stubs Stub2 is open to the odd harmonic waves and closed to the even harmonic waves, which is useful to prevent the leakage of energy from the bias network and promote the linearity and output power of the amplifier. It can also improve the microwave performance of the power device in output power and gain [10].

An important effort for input matching was required to control the gain flatness over 3 to 8 GHz. An open radial stubs Stub1 with the characteristic of broadband impedance transformer is used during the multi impedance matching design. R1 and R2 are used to achieve required stability within the band of interest. But improved stability and better VSWR are achieved at a cost of gain. Therefore, to compensate the gain, a parallel capacitor C2 is used along with the series resistor R1. This arrangement is called hybrid stability network which makes it possible to attain stability and broadband matching simultaneously for R1 and C2 respectively tuning the gain of low and high frequency signal in the working band [11]. The resistor R2 between DC power and the transistor gate for the GaN HEMT

neither consumes any power nor produces any heat, because the gate current is nearly zero under ideal conditions. The input and output coupling capacitors are shown as C1 and C3 respectively, which can insulate the DC current and match simultaneously.

4 Results and discussion

The power amplifier module is shown in Figure 4. Resistors and capacitors are connected by silver paste. The bonding wires are used to connect the GaN HEMT with the micro strip on the PCB. And a Mo-Cu plate which thermal expansion coefficient is between SiC (substrate of GaN HEMT) and metal copper (as a heat sink) is used as the chip carrier. What's more, because the GaN HEMT is a depleted field effect transistor, this transistor must be supplied by a double power supply. And a negative voltage of the gate must be outputted at first and then a positive voltage of the drain is outputted. The negative voltage switches off after positive voltage. This can ensure the transistor works safely. The power amplifier circuit is measured under DC bias of $V_{ds}=28$ V, $V_{gs}=-2.53$ V.



Figure 4 Image of the fabricated wideband amplifier

Figure 5 shows the measured and simulated S parameter of the circuit. These obtained performances are very promising and have a further improvement than before [9]. The measured S (2, 1) is more than 11 dB in the frequency from 3 to 8 GHz. The measured input return loss S (1, 1) is less than -6.5 dB which is a little poorer than that of the simulation. The measured output return loss S (2, 2) shows great performance and -9.8 to 24.4 dB is measured from 3 to 8 GHz. Although the measured S (1, 1) and S (2, 2) have some frequency deviation, they have the same change trend as those of the simulation and show good spread behaviour.

Figure 6 shows the measured and simulated CW output power and power gain at the frequency of 5 GHz. It shows that the measured output power is 31.8 dBm at 1dB gain compression and a measured maximum output power of 37.3 dBm. The measured output power and power gain are a little less than ones of the simulation. One important reason is that the heat produced by the chip under large signal operations can't be well dissipated. Better heat conduction material should be used. In addition, the established model has some errors,

which also lead to some differences between simulated and measured results.

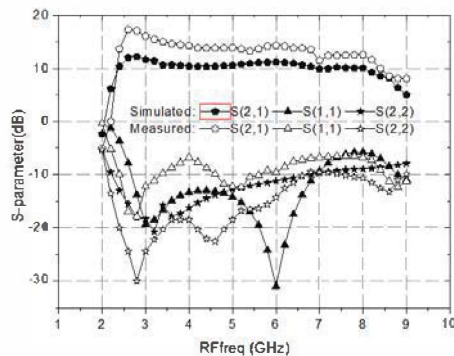


Figure 5 Simulated and measured S-parameter of the circuit

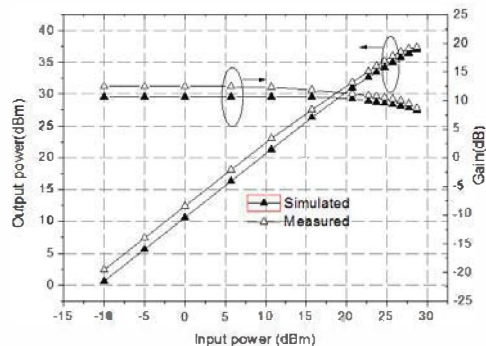


Figure 6 Output power and gain versus input power at 5 GHz

5 Conclusions

An ultra-wideband power amplifier is designed using the wideband matching networks for frequency ranging from 3-8 GHz. The amplifier was fabricated and high power measurements are performed in CW operations. It has been shown that the fabricated amplifier provides a good stability, high gain and better input & output return losses. The measured $S(2, 1)$ is more than 11 dB and the saturated output power is 37.3 dBm. The results demonstrate the effectiveness of the adopted design approach. All these performance are promising for the future development on ultra-wideband power amplifier.

Acknowledgements

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